DitaCraft User Guide

The Complete Guide to DITA Editing in VS Code

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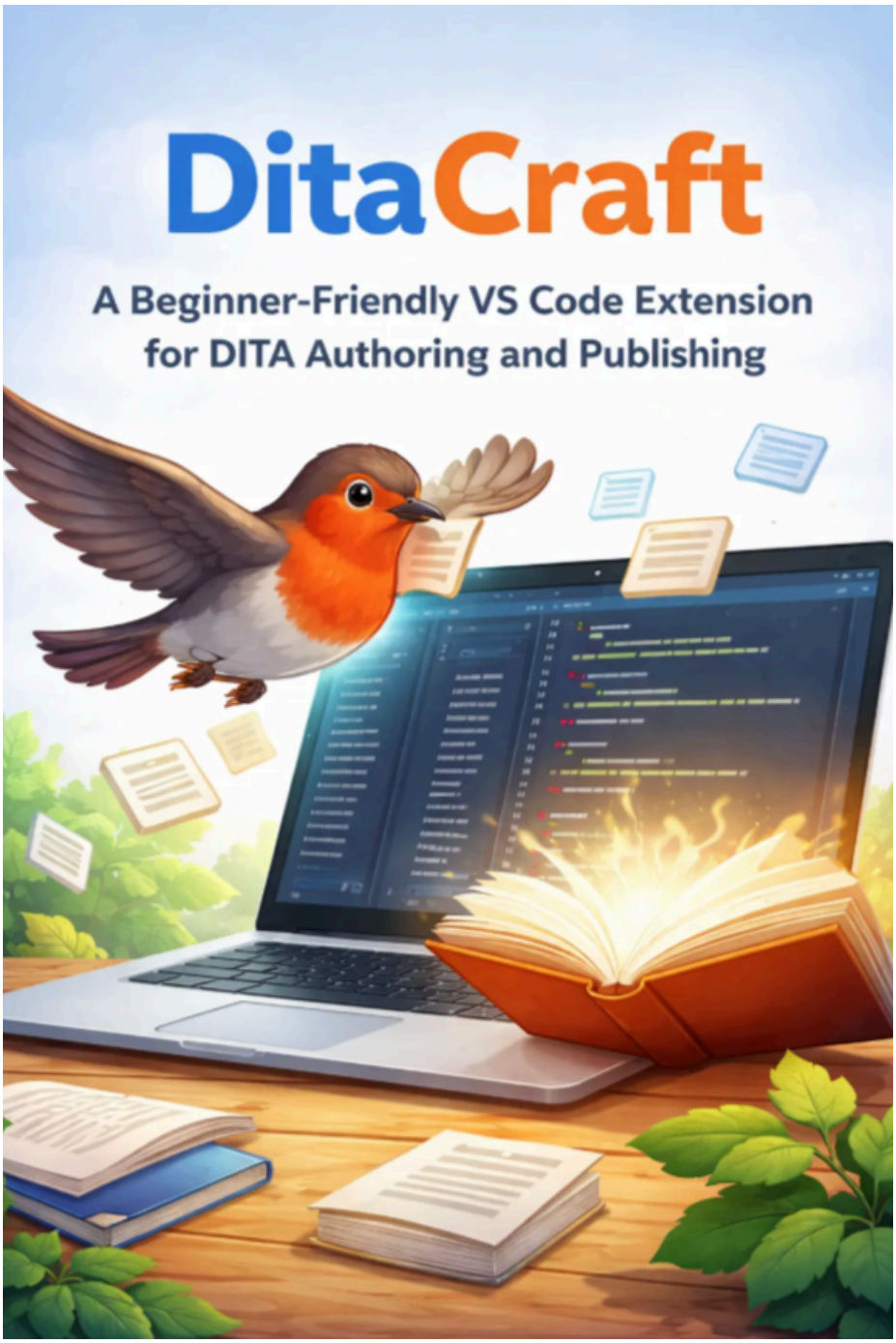

DitaCraft User Guide

The Complete Guide to DITA Editing in VS Code

Version 0.6.2

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DitaCraft Project

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Preface

Preface

Welcome to the DitaCraft User Guide.

About This Guide

This guide provides comprehensive documentation for DitaCraft, the Visual Studio Code extension for DITA authoring and publishing. Whether you are new to DITA or an experienced technical writer, this guide will help you get the most out of DitaCraft.

Intended Audience

This guide is intended for:

- **Technical Writers** who create and maintain DITA documentation
- **Documentation Managers** who oversee DITA projects
- **Developers** who write technical documentation alongside code
- **Content Strategists** evaluating DITA tooling options

Prerequisites

To use DitaCraft effectively, you should have:

- Visual Studio Code 1.80 or higher installed
- Basic familiarity with XML concepts
- Understanding of DITA fundamentals (helpful but not required)

How to Use This Guide

This guide is organized into the following sections:

Part I: Getting Started	Introduction to DitaCraft and installation steps
Part II: Using DitaCraft	Commands reference and detailed feature explanations
Part III: Configuration	Settings and customization options
Appendix: Keyboard Shortcuts	Quick reference for all keyboard shortcuts
Glossary	Definitions of key terms used throughout the guide

Conventions Used

This guide uses the following conventions:

Convention	Meaning
Bold text	UI elements, buttons, menu items
`Monospace`	Code, commands, file names, settings
`File paths`	Directory and file paths
`User input`	Text you should type

Version Information

This guide documents DitaCraft version 0.6.2. For the latest updates and release notes, visit the project repository.

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Part

I

Getting Started

Topics:

- Introduction to DitaCraft
- Getting Started

Chapter 1

Introduction to DitaCraft

DitaCraft is a comprehensive Visual Studio Code extension for editing and publishing DITA content.

What is DitaCraft?

DitaCraft is the easiest way to edit and publish your DITA (Darwin Information Typing Architecture) files directly from Visual Studio Code. It provides a complete authoring environment with:

- LSP-powered IntelliSense with auto-completion, hover docs, and document symbols
- Full DTD validation with three engine options
- Smart navigation with Ctrl+Click on href, conref, keyref, and conkeyref
- Cross-file Go to Definition, Find References, and Rename
- Key space resolution for keyref and conkeyref with key scope support
- 35 DITA validation rules with DITA 2.0 awareness
- Localized diagnostics (English, French)
- Explicit root map selection with status bar indicator
- DITAVAL conditional processing support
- Live HTML5 preview with bidirectional scroll sync
- One-click publishing via DITA-OT
- Interactive map visualizer
- XML formatting with code actions and quick fixes

Key Features

DitaCraft offers the following key features:

LSP-Powered IntelliSense	Context-aware auto-completion for DITA elements, attributes, and values. Hover documentation for 364+ elements. Document outline and workspace symbol search.
Smart Navigation	Ctrl+Click on `href`, `conref`, `keyref`, and `conkeyref` attributes to navigate directly to referenced content. Cross-file Go to Definition, Find References, and Rename.
Full DTD Validation	Real-time validation with three engine options: built-in (default), TypesXML (full DTD), and xmllint.

	LSP-powered diagnostics for instant feedback as you type.
Live Preview	Side-by-side HTML5 preview with bidirectional scroll sync, theme support, and print mode.
DITA-OT Integration	Publish to HTML5, PDF, EPUB, and more with direct DITA Open Toolkit integration.
DITAVAL Support	Full IntelliSense for DITAVAL conditional processing files including element completion, attribute values, and action keywords.
Map Visualizer	Interactive tree view showing your DITA map hierarchy with navigation support.

Supported File Types

DitaCraft supports the following DITA file types:

Extension	Description
`.dita`	DITA topics (concept, task, reference, topic)
`.ditamap`	DITA maps for organizing topics
`.bookmap`	Book-oriented maps with chapters and parts
`.ditaval`	DITAVAL conditional processing profiles

Version Information

This guide covers DitaCraft version 0.6.2, which includes:

- LSP server with IntelliSense, diagnostics, and navigation
- Cross-reference validation for href, conref, keyref, and conkeyref
- DITA Rules Engine with 35 Schematron-equivalent rules (including 10 DITA 2.0-specific rules)
- Profiling validation with subject scheme support and hierarchy grouping
- Activity Bar views: DITA Explorer, Key Space, and Diagnostics
- Explicit root map selection with status bar indicator
- Localized diagnostics (English and French)
- RNG (RelaxNG) and OASIS Catalog validation services
- Conref/conkeyref content preview on hover
- 9 code actions including accessibility quick fixes
- DITAVAL conditional processing support
- DITA version detection and error-tolerant XML tokenizer
- Smart debouncing (300ms topics, 1000ms maps) with per-document cancellation
- Key scope support with `@keyscope` attribute handling
- 1040+ comprehensive tests (620 client + 419 server)

Chapter 2

Getting Started

Learn how to install and configure DitaCraft for your first DITA project.

Before you begin

Before installing DitaCraft, ensure you have:

- Visual Studio Code 1.80 or higher
- DITA-OT 4.2.1 or higher (for publishing features)

About this task

This guide walks you through installing DitaCraft and creating your first DITA file.

Procedure

1. Install DitaCraft from VS Code Marketplace
   a) Open VS Code
   b) Press **Ctrl+P** (or **Cmd+P** on Mac)
   c) Type `ext install ditacraft` and press Enter
2. Configure DITA-OT path (optional, required for publishing)
   a) Download DITA-OT from dita-ot.org
   b) Extract to a location (e.g., `C:\DITA-OT-4.2.1`)
   c) Open Command Palette (**Ctrl+Shift+P**)
   d) Type `DITA: Configure DITA-OT Path`
   e) Select your DITA-OT installation directory
3. Create your first DITA topic
   a) Open Command Palette (**Ctrl+Shift+P**)
   b) Type `DITA: Create New Topic`
   c) Select topic type (concept, task, or reference)
   d) Enter a file name for your topic

   A new DITA topic file is created with proper DOCTYPE and structure.
4. Validate your DITA file
   a) Open your DITA file
   b) Press **Ctrl+Shift+V** to validate

   Validation results appear in the Problems panel. Errors and warnings are highlighted in the editor.
5. Preview your content (optional)
   a) Press **Ctrl+Shift+H** to open HTML5 preview

   A side-by-side preview panel opens showing your rendered DITA content.

Results

You now have DitaCraft installed and configured. You can:

- Create DITA topics, maps, and bookmaps
- Validate your content in real-time
- Navigate between files using Ctrl+Click
- Preview and publish your documentation

Part II

Using DitaCraft

Topics:

- Commands Overview
- Validation Commands
- Publishing Commands
- File Creation Commands
- Navigation Commands
- Features Overview
- IntelliSense
- Smart Navigation
- Validation
- Live Preview
- Map Visualizer
- Key Resolution
- DITAVAL Support
- Cross-Reference Validation
- DITA Rules Engine
- Profiling and Subject Scheme Validation
- Activity Bar Views
- DITA Explorer
- Key Space View
- Diagnostics View
- Root Map Management
- Localization (i18n)

Chapter 3

Commands Overview

DitaCraft provides commands for validation, publishing, file creation, and navigation.

Accessing Commands

All DitaCraft commands are accessible via the Command Palette:

1. Press **Ctrl+Shift+P** (or **Cmd+Shift+P** on Mac)
2. Type `DITA:` to see all available commands
3. Select the desired command

Command Categories

DitaCraft commands are organized into the following categories:

Validation Commands	Validate DITA files for XML syntax and DTD conformance
Publishing Commands	Publish DITA content to HTML5, PDF, and other formats
File Creation Commands	Create new DITA topics, maps, and bookmaps
Navigation Commands	Navigate and visualize DITA content structure

Quick Command Reference

Command	Shortcut	Description
DITA: Validate Current File	Ctrl+Shift+V	Validate the active DITA file
DITA: Publish (Select Format)	Ctrl+Shift+B	Publish with format selection
DITA: Publish to HTML5	-	Quick publish to HTML5
DITA: Preview HTML5	Ctrl+Shift+H	Show live HTML5 preview
DITA: Show Map Visualizer	-	Display interactive map hierarchy
DITA: Create New Topic	-	Create a new DITA topic

Command	Shortcut	Description
DITA: Create New Map	-	Create a new DITA map
DITA: Create New Bookmap	-	Create a new bookmap
DITA: Configure DITA-OT Path	-	Set DITA-OT installation path
DITA: Set Root Map	-	Set an explicit root map for key resolution
DITA: Clear Root Map (Auto-Discover)	-	Return to automatic root map discovery
DITA: Open File	-	Open a file from the DITA Explorer tree

Chapter 4

Validation Commands

Commands for validating DITA files against XML syntax and DTD specifications.

DITA: Validate Current File

Shortcut: Ctrl+Shift+V (Windows/Linux) or **Cmd+Shift+V** (Mac)

Validates the currently active DITA file for:

- XML well-formedness (proper tags, attributes, encoding)
- DTD conformance (valid elements, required attributes)
- DITA structure (required elements like title, proper nesting)
- Content model validation (correct child elements)

Usage:

1. Open a `.dita`, `.ditamap`, or `.bookmap` file
2. Press **Ctrl+Shift+V** or run the command from Command Palette
3. View results in the Problems panel

Output:

- **Errors** - Critical issues that must be fixed (red squiggles)
- **Warnings** - Potential issues to review (yellow squiggles)
- **Information** - Suggestions and notes (blue squiggles)

Automatic Validation

DitaCraft automatically validates DITA files when:

- **On Open:** When you open a DITA file
- **On Save:** When you save a DITA file (if `autoValidate` is enabled)
- **On Change:** After you stop typing (with 500ms debounce)

To disable automatic validation, set `ditacraft.autoValidate` to `false` in settings.

Validation Engines

DitaCraft supports three validation engines:

Engine	Description	Requirements
`typesxml`	Full DTD validation with 100% W3C conformance (default, recommended)	None (bundled)

Engine	Description	Requirements
`built-in`	Lightweight XML and content model checking	None (bundled)
`xmllint`	External validation using libxml2	xmllint installed on system

To change the validation engine, set `ditacraft.validationEngine` in settings.

Rate Limiting

To prevent performance issues from rapid validation requests, DitaCraft includes rate limiting:

- Maximum 10 validation requests per second per file
- Excess requests are silently skipped (auto-validation) or show a warning (manual validation)
- Rate limits reset after the time window expires

Chapter 5

Publishing Commands

Commands for publishing DITA content to HTML5, PDF, and other output formats.

DITA: Publish (Select Format)

Shortcut: Ctrl+Shift+B (Windows/Linux) or **Cmd+Shift+B** (Mac)

Opens a format selection dialog and publishes the current DITA map or topic.

Available Formats:

- **html5** - Modern HTML5 output with responsive design
- **pdf** - PDF output (requires Apache FOP or similar)
- **xhtml** - XHTML 1.0 output
- **eclipsehelp** - Eclipse help system format
- **htmlhelp** - Microsoft HTML Help format
- **markdown** - Markdown output

Requirements:

- DITA-OT must be installed and configured
- Java Runtime Environment (JRE) 11 or higher

DITA: Publish to HTML5

Quickly publishes the current file to HTML5 format without format selection.

Usage:

1. Open a `.ditamap` or `.bookmap` file
2. Run the command from Command Palette
3. Output is generated in the `out` folder

Output Location:

By default, output is generated in `{workspace}/out/{format}`. You can customize this in settings with `ditacraft.outputDirectory`.

DITA: Preview HTML5

Shortcut: Ctrl+Shift+H (Windows/Linux) or **Cmd+Shift+H** (Mac)

Opens a live HTML5 preview panel alongside the editor.

Features:

- **Live Updates:** Preview refreshes automatically as you type
- **Scroll Sync:** Bidirectional scroll synchronization between editor and preview
- **Theme Support:** Light and dark theme support

- **Print Mode:** Toggle print-friendly styling
- **Conref Resolution:** Displays resolved conref content

Tip: The preview uses a lightweight renderer that does not require DITA-OT, making it instant for quick content review.

DITA: Configure DITA-OT Path

Opens a folder picker to select your DITA-OT installation directory.

Steps:

1. Download DITA-OT from dita-ot.org
2. Extract to a location on your system
3. Run this command and select the DITA-OT directory

Verification:

After configuration, DitaCraft verifies the installation by checking for:

- `bin/dita` or `bin/dita.bat` executable
- `lib/` directory with required JAR files
- Valid version information

Common Publishing Errors

If publishing fails, check the following:

DITA-OT not found	Run **DITA: Configure DITA-OT Path** to set the correct path.
Java not found	Install JRE 11+ and ensure `JAVA_HOME` is set correctly.
Build failed	Check the Output panel for detailed error messages from DITA-OT.

Chapter 6

File Creation Commands

Commands for creating new DITA topics, maps, and bookmaps.

DITA: Create New Topic

Creates a new DITA topic file with proper DOCTYPE and structure.

Topic Types:

Concept	For conceptual information that helps users understand a subject. Uses `<concept>` root element with `<conbody>`.
Task	For procedural information with step-by-step instructions. Uses `<task>` root element with `<taskbody>` and `<steps>`.
Reference	For reference information like API documentation or command syntax. Uses `<reference>` root element with `<refbody>`.
Topic	Generic topic type for content that doesn't fit other categories. Uses `<topic>` root element with `<body>`.

Usage:

1. Run the command from Command Palette
2. Select the topic type
3. Enter a file name (without extension)
4. The file is created in the current folder with `.dita` extension

DITA: Create New Map

Creates a new DITA map file for organizing topics.

Map Structure:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE map PUBLIC "-//OASIS//DTD DITA Map//EN"
 "map.dtd">
<map id="my-map">
    <title>My Map Title</title>
    <topicref href="topic1.dita"/>
    <topicref href="topic2.dita"/>
```

```
</map>
```

Usage:

1. Run the command from Command Palette
2. Enter a file name (without extension)
3. The file is created with `.ditamap` extension

After creation, add `<topicref>` elements to reference your topics.

DITA: Create New Bookmap

Creates a new bookmap for book-oriented publications.

Bookmap Structure:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE bookmap PUBLIC "-//OASIS//DTD DITA
 BookMap//EN" "bookmap.dtd">
<bookmap id="my-bookmap">
    <booktitle>
        <mainbooktitle>My Book Title</
mainbooktitle>
    </booktitle>
    <frontmatter>
        <booklists><toc/></booklists>
    </frontmatter>
    <chapter href="chapter1.dita"/>
    <backmatter>
        <booklists><indexlist/></booklists>
    </backmatter>
</bookmap>
```

Bookmap Elements:

- `<booktitle>` - Book title information
- `<bookmeta>` - Metadata (author, dates, rights)
- `<frontmatter>` - TOC, preface, dedication
- `<chapter>` - Main content chapters
- `<part>` - Groups of chapters
- `<appendix>` - Supplementary content
- `<backmatter>` - Index, glossary

File Naming Best Practices

- Use lowercase letters and hyphens (e.g., `getting-started.dita`)
- Avoid spaces and special characters
- Use descriptive names that reflect content
- Keep names concise but meaningful

Chapter 7

Navigation Commands

Commands for navigating and visualizing DITA content structure.

DITA: Show Map Visualizer

Opens an interactive tree view showing your DITA map hierarchy.

Features:

- **Hierarchical View:** Displays maps, chapters, and topics in a tree structure
- **Click Navigation:** Click any node to open the corresponding file
- **Real-time Updates:** Tree updates when files are added or removed
- **Status Indicators:** Shows validation status for each file

Usage:

1. Open a `.ditamap` or `.bookmap` file
2. Run the command from Command Palette
3. The visualizer opens in a side panel

Ctrl+Click Navigation

DitaCraft provides smart navigation using Ctrl+Click (Cmd+Click on Mac) on various attributes.

Supported Attributes:

Attribute	Navigation Behavior
`href`	Opens the referenced file
`conref`	Navigates to the content reference target (file#id)
`keyref`	Resolves the key and navigates to its target
`conkeyref`	Resolves the key and navigates to the content reference

Example:

```
<topicref href="topics/intro.dita"/>  <!-- Ctrl
+Click opens intro.dita -->
<p conref="shared.dita#shared/note1"/>  <!-- Ctrl
+Click opens shared.dita at #note1 -->
<ph keyref="product-name"/>  <!-- Ctrl+Click
 resolves key and navigates -->
```

Go to Definition

Press **F12** or right-click and select **Go to Definition** to navigate to referenced content.

This works on the same attributes as Ctrl+Click navigation.

Peek Definition

Press **Alt+F12** to peek at the definition without leaving your current file.

A small inline editor shows the referenced content, allowing you to:

- View the target content
- Make quick edits
- Navigate to the full file

Find All References

Press **Shift+F12** to find all places where the current element or ID is referenced.

Useful for:

- Finding all conrefs to a specific element
- Locating topic references in maps
- Understanding content reuse patterns

Chapter 8

Features Overview

DitaCraft provides a comprehensive set of features for DITA authoring in Visual Studio Code.

Core Features

DitaCraft transforms VS Code into a full-featured DITA authoring environment with the following capabilities:

LSP-Powered IntelliSense	Context-aware auto-completion for 364+ DITA elements, attributes, and attribute values. Hover documentation, document outline, and workspace-wide symbol search (**Ctrl+T**). Powered by a built-in Language Server Protocol implementation.
Smart Navigation	Navigate between files using Ctrl +Click on href, conref, keyref, and conkeyref attributes. Cross-file Go to Definition, Find References, and Rename for IDs and references.
Full DTD Validation	Real-time validation with three engine options: built-in (default, lightweight), TypesXML (full DTD, 100% W3C conformance), and xmllint (external). LSP diagnostics provide instant feedback as you type.
DITAVAL Support	Full IntelliSense for DITAVAL conditional processing files with element completion, attribute suggestions, and action keyword values.
Live Preview	Side-by-side HTML5 preview with bidirectional scroll sync. See your formatted output without running a full build.
DITA-OT Integration	One-click publishing to HTML5, PDF, and other formats using DITA Open Toolkit integration.

Map Visualizer	Interactive tree view showing your DITA map hierarchy with click-to-navigate functionality.
Key Resolution	Full key space support for keyref and conkeyref resolution with improved root map discovery and intelligent key definition lookup.
Cross-Reference Validation	Validates href, conref, keyref, and conkeyref references across your workspace. Catches broken links, missing target files, and undefined keys before publishing.
DITA Rules Engine	35 Schematron-equivalent rules organized in four categories (mandatory, recommendation, authoring, accessibility) enforce DITA best practices and specification requirements. Includes 10 DITA 2.0-specific rules for removed elements and attributes, with automatic version detection.
Profiling and Subject Scheme Validation	Validates profiling attribute values (audience, platform, product, etc.) against subject scheme constraints, ensuring controlled vocabularies are respected.
Activity Bar Views	Dedicated DitaCraft sidebar with three tree views: DITA Explorer (map hierarchy browser), Key Space (defined/undefined/unused keys), and Diagnostics (validation issues grouped by file or severity). Includes file decoration badges for validation status and root map selector in the status bar.
Root Map Management	Explicitly set a root map for your project to control key resolution, cross-reference validation, and subject scheme discovery. Status bar indicator shows the current root map. See Root Map Management on page 71 for details.
Localized Diagnostics	All diagnostic messages are localized with support for English and French. The language follows your VS Code display language setting automatically. See Localization (i18n) on page 73 for details.

Editor Enhancements

DitaCraft enhances the VS Code editing experience for DITA files:

- **Syntax Highlighting:** DITA-aware XML syntax highlighting
- **Auto-completion:** Context-aware element, attribute, and value suggestions via LSP
- **Hover Documentation:** Hover over any DITA element to see its description and allowed attributes. Hover over keyref/conkeyref to preview resolved content inline
- **Linked Editing:** Edit opening and closing XML tags simultaneously
- **Bracket Matching:** Automatic tag matching and highlighting
- **Folding:** Collapse sections, steps, and other block elements
- **Outline View:** Navigate document structure in the Outline panel
- **Workspace Symbols:** Search for elements across all open files with **Ctrl +T**
- **Code Actions:** Quick fixes for missing DOCTYPE, missing ID, missing title, empty elements, duplicate IDs, DITA rules violations (add `@otherrole`, remove deprecated `<indextermref>`, convert `alt` attribute to `<alt>` element, add missing `<alt>`)
- **XML Formatting:** Format DITA documents with proper indentation and inline/block handling

Validation Features

DitaCraft offers multiple validation options:

- **Real-time Validation:** LSP-powered diagnostics appear as you type with smart debouncing (300ms for topics, 1000ms for maps)
- **Multiple Engines:** Choose from built-in (default), TypesXML, or xmllint, with optional RNG (RelaxNG) validation
- **OASIS Catalog Support:** DTD validation via TypesXML with OASIS XML Catalog resolution and parser pool optimization
- **Detailed Diagnostics:** Line-precise error messages with fix suggestions, localized in English and French
- **Code Actions:** Quick fixes for common issues (missing DOCTYPE, IDs, titles) and DITA rules violations
- **Cross-Reference Validation:** Validates href, conref, keyref, and conkeyref targets across files
- **DITA Rules Engine:** 35 Schematron-equivalent rules including 10 DITA 2.0-specific rules
- **Profiling Validation:** Subject scheme-based validation with hierarchy grouping and default value preselection
- **Rate Limiting:** Protection against excessive validation requests

Productivity Features

- **Quick File Creation:** Create topics, maps, and bookmaps from Command Palette
- **Content Reuse:** Full support for conref and keyref mechanisms
- **Cross-file References:** Find all references to an element across your entire workspace
- **Cross-file Rename:** Rename an ID and update all references across files
- **Document Links:** Clickable links for href, conref, keyref, and conkeyref attributes
- **Workspace Support:** Works with single files or multi-root workspaces
- **cSpell Integration:** Auto-prompted spell checking setup with DITA-specific dictionary

- **Activity Bar Views:** Dedicated sidebar with DITA Explorer, Key Space browser, and Diagnostics view for project-level navigation and issue tracking

Chapter 9

IntelliSense

DitaCraft provides LSP-powered IntelliSense for DITA authoring with auto-completion, hover documentation, and symbol navigation.

Overview

DitaCraft includes a built-in Language Server Protocol (LSP) implementation that provides rich editing features for DITA files. The LSP server runs in the background and provides real-time feedback as you type, including auto-completion suggestions, hover documentation, and document symbols.

Auto-Completion

DitaCraft offers context-aware auto-completion for DITA content:

Element Completion	Type < to see a list of valid child elements for the current context. The suggestions are filtered based on the parent element's content model, showing only elements that are allowed at the current position. Supports 364+ DITA elements.
Attribute Completion	Inside an opening tag, type a space to see available attributes for the current element. Common attributes like `id`, `href`, and `class` are suggested based on the element type.
Attribute Value Completion	When typing an attribute value (after =), DitaCraft suggests valid values from predefined enumerations. For example, the `type` attribute on `<note>` suggests values like `attention`, `caution`, `danger`, etc.
Subject Scheme Completions	When a subject scheme map constrains profiling attributes, auto-completion shows only allowed values with hierarchy grouping (e.g., `Platform > Linux > Ubuntu`). Default values from the scheme are preselected in the completion list.

Hover Documentation

Hover over any DITA element name to see its documentation, including:

- A description of the element's purpose
- Common attributes and their usage
- The element's content model (what children are allowed)

Documentation is available for 142+ DITA elements covering all commonly used topic types, structural elements, and inline elements.

Reference Preview on Hover: Hover over key references and content references to see resolved content inline:

- **keyref hover:** Shows the key's target file, navtitle, short description, and inline content
- **conkeyref hover:** Resolves the key, then extracts and displays a preview of the referenced element content as XML
- **href/conref hover:** Shows target file information and navigation path

Document Symbols

DitaCraft provides document symbols for navigating your DITA content:

- **Outline View:** The VS Code Outline panel shows the structure of your DITA document with sections, steps, tables, and other structural elements.
- **Go to Symbol:** Press **Ctrl+Shift+O** to quickly navigate to any element in the current document.
- **Workspace Symbols:** Press **Ctrl+T** to search for elements across all DITA files in your workspace.

Code Actions and Quick Fixes

DitaCraft provides code actions (quick fixes) for common issues. When the lightbulb icon appears, click it or press **Ctrl+.** to see available fixes:

- **Add Missing DOCTYPE:** Automatically adds the correct DOCTYPE declaration for your topic type
- **Add Missing ID:** Generates a unique ID attribute for the root element
- **Add Missing Title:** Inserts a `<title>` element where required
- **Remove Empty Element:** Cleans up empty elements that have no content
- **Fix Duplicate ID:** Offers to rename a duplicate ID to a unique value
- **Add `@otherrole`:** Inserts the missing `otherrole` attribute when `role="other"` is detected
- **Remove Deprecated `<indextermref>`:** Removes the deprecated element flagged by the DITA Rules Engine
- **Convert `alt` Attribute to `<alt>` Element:** Migrates the deprecated `alt` attribute syntax on `<image>` to the modern `<alt>` child element
- **Add Missing `<alt>`:** Inserts an empty `<alt>` child element inside `<image>` for accessibility compliance

Linked Editing

When you edit an XML tag name, DitaCraft automatically updates the matching opening or closing tag. This feature uses linked editing ranges to keep tag pairs synchronized.

For example, if you change `<section>` to `<example>`, the corresponding `</section>` is automatically updated to `</example>`.

XML Formatting

DitaCraft formats DITA XML documents with proper indentation and element handling:

- **Block Elements:** Structural elements like `<section>`, `<p>`, `<ul>` are formatted with newlines and indentation
- **Inline Elements:** Elements like `<b>`, `<codeph>`, `<xref>` are kept inline with surrounding text
- **Preformatted Content:** `<codeblock>` and `<pre>` content is preserved as-is

Format a document with **Shift+Alt+F** or format a selection with **Ctrl+K Ctrl +F**.

Chapter 10

Smart Navigation

Navigate between DITA files using Ctrl+Click on reference attributes.

Overview

Smart Navigation allows you to quickly move between related DITA files by clicking on reference attributes. Hold **Ctrl** (or **Cmd** on Mac) and click on any supported attribute value to navigate to the target.

href Navigation

The `href` attribute is used in maps to reference topics and in topics to link to other content.

Examples:

```
<!-- In a ditamap -->
<topicref href="topics/getting-started.dita"/>

<!-- In a topic -->
<xref href="reference-guide.dita#config-settings"/
>
```

Ctrl+Click on the href value opens the referenced file. If the href includes a fragment identifier (#id), the cursor moves to that element.

conref Navigation

Content references (`conref`) pull content from one location to another. Navigation takes you to the source element.

Example:

```
<p conref="shared/warnings.dita#warnings/safety-
note"/>
```

Ctrl+Click navigates to `shared/warnings.dita` and positions the cursor at the element with `id="safety-note"`.

keyref Navigation

Key references use indirect addressing through keys defined in a map.

Map Definition:

```
<keydef keys="product-name">
    <topicmeta>
        <keywords><keyword>DitaCraft</keyword></
keywords>
```

```
    </topicmeta>
</keydef>

<keydef keys="install-guide" href="topics/
installation.dita"/>
```

Usage:

```
<ph keyref="product-name"/>  <!-- Resolves to
 "DitaCraft" -->
<xref keyref="install-guide"/>  <!-- Links to
 installation.dita -->
```

Ctrl+Click on a keyref:

1. Resolves the key using the current key space
2. Navigates to the key's target (href or definition)

conkeyref Navigation

Content key references combine key resolution with content referencing.

Example:

```
<!-- Key definition -->
<keydef keys="shared-content" href="shared/
common.dita"/>

<!-- Usage -->
<note conkeyref="shared-content/warning-note"/>
```

Ctrl+Click resolves `shared-content` to `shared/common.dita`, then navigates to `#warning-note`.

Cross-File Features

DitaCraft provides cross-file navigation and editing via the LSP server:

Go to Definition	Press **F12** or Ctrl+Click on href, conref, keyref, or conkeyref to jump to the target element across files.
Find All References	Press **Shift+F12** on an `id` attribute to find all conref and href references to that element across your workspace.
Rename Symbol	Press **F2** on an `id` attribute to rename it and update all references across files.
Document Links	All href, conref, keyref, and conkeyref values are clickable links. Hover to see the resolved target path.

Navigation Tips

- **Hover Preview:** Hover over a reference to see a preview without navigating
- **Go Back:** Use **Alt+Left Arrow** to return to the previous location
- **Peek Definition:** Use **Alt+F12** to peek without leaving your file

- **Open to Side: Ctrl+Alt+Click** opens the target in a split editor

Chapter 11

Validation

DitaCraft validates DITA files against XML syntax, DTD specifications, cross-references, DITA rules, and profiling constraints in real-time.

Validation Overview

DitaCraft provides comprehensive validation to ensure your DITA content is well-formed, conforms to DITA specifications, uses valid cross-references, follows DITA best practices, and respects profiling constraints. Validation runs automatically as you edit and can also be triggered manually.

What Gets Validated

XML Well-formedness	Checks for proper XML syntax: matching tags, valid attribute values, proper encoding, and correct entity references.
DTD Conformance	Validates elements and attributes against DITA 1.3 DTDs. Ensures elements appear in allowed contexts with required attributes.
Content Model	Verifies child elements appear in the correct order and quantity according to the DTD content model.
ID Uniqueness	Checks that ID attributes are unique within a topic.
Cross-Reference Validation	Validates href, conref, keyref, and conkeyref attributes to ensure target files exist, topic and element IDs are present, and keys are defined in the key space. See Cross-Reference Validation on page 55 for details.
DITA Rules Engine	35 Schematron-equivalent rules in four categories (mandatory, recommendation, authoring, accessibility) enforce DITA specification requirements and best practices. Includes 10 DITA 2.0-specific rules for removed elements and attributes. Rules are version-aware and include quick fixes. See DITA Rules Engine on page 57 for details.

Profiling Validation	When a subject scheme map is present, validates that profiling attribute values (audience, platform, product, etc.) use only the allowed values defined by the scheme. See Profiling and Subject Scheme Validation on page 61 for details.

Validation Engines

DitaCraft supports three validation engines, each with different capabilities:

Engine	Pros	Cons
`typesxml`	Full DTD support, 100% W3C conformance, no external dependencies	Slightly slower for very large files
`built-in`	Very fast, lightweight, basic content model checking	Limited DTD support, may miss some errors
`xmllint`	Industry-standard libxml2, excellent performance	Requires external installation

Recommendation: The `built-in` engine (the default) provides a good balance of speed and accuracy for most users. Switch to `typesxml` when you need strict DTD conformance checking.

Additional Validation:

- **OASIS Catalog Support:** The `typesxml` engine uses an OASIS XML Catalog to resolve PUBLIC identifiers to bundled DTDs, with a parser pool (3 instances) for efficient reuse.
- **RNG (RelaxNG) Validation:** Optional RelaxNG schema validation via `salve-annos` and `saxes`. Enable by setting `ditacraft.schemaFormat` to `"rng"` and providing a schema path via `ditacraft.rngSchemaPath`. RNG schemas are compiled to JSON and cached (up to 20 grammars).

LSP-Powered Diagnostics

DitaCraft uses a built-in Language Server Protocol (LSP) implementation to provide real-time diagnostics. The LSP server validates your documents in the background and reports issues instantly as you type, without waiting for a manual save or validation trigger.

The LSP diagnostics cover:

- XML well-formedness (mismatched tags, invalid syntax)
- DITA structure validation (element context, required attributes)
- ID uniqueness within documents
- Cross-reference validation (href, conref, keyref, conkeyref)
- DITA rules (specification compliance, best practices, accessibility)
- Profiling validation (subject scheme constraints)

LSP diagnostics appear with the source label `dita-lsp` in the Problems panel, while client-side validations appear with the `dita` source label. DITA rules diagnostics appear with the source label `dita-rules`.

All diagnostic messages are localized — see Localization (i18n) on page 73 for supported languages.

Diagnostic Levels

Validation issues are reported at three severity levels:

- **Error (Red):** Critical issues that must be fixed. The document is not valid DITA.
- **Warning (Yellow):** Potential issues that should be reviewed. The document may be valid but has problems.
- **Information (Blue):** Suggestions and notes. Not errors, but helpful for improvement.

Viewing Validation Results

Validation results appear in multiple places:

- **Editor Squiggles:** Colored underlines appear on problematic content
- **Problems Panel:** Full list of all diagnostics (**Ctrl+Shift+M**)
- **Status Bar:** Shows error/warning count for the current file
- **File Explorer:** File icons show validation status

Automatic Validation

By default, DitaCraft validates automatically:

- **On Open:** When you open a DITA file
- **On Save:** When you save a file (configurable)
- **On Change:** After you stop typing, with smart debouncing:
  - **Topics:** 300ms debounce for fast feedback
  - **Maps:** 1000ms debounce (maps are more expensive to validate)

  Per-document cancellation ensures only the latest edit triggers validation.

To disable automatic validation, set `ditacraft.autoValidate` to `false` in settings.

Rate Limiting

To prevent performance issues, DitaCraft limits validation requests:

- Maximum 10 requests per second per file
- Excess requests are silently skipped during auto-validation
- Manual validation shows a warning if rate limited

This protects against excessive CPU usage when making rapid edits.

Chapter 12

Live Preview

View your DITA content as formatted HTML5 in a side-by-side preview panel.

Overview

The Live Preview feature renders your DITA content as HTML5 in real-time, allowing you to see how your documentation will look without running a full DITA-OT build.

Shortcut: Ctrl+Shift+H (Windows/Linux) or **Cmd+Shift+H** (Mac)

Preview Features

Live Updates	The preview refreshes automatically as you type. Changes appear within milliseconds, providing instant feedback on your edits.
Bidirectional Scroll Sync	Scrolling in the editor scrolls the preview to the corresponding position, and vice versa. Keep your place when reviewing long documents.
Theme Support	The preview adapts to VS Code's current theme. Light themes show light preview; dark themes show dark preview.
Print Mode	Toggle print mode to see how your content will appear when printed. Useful for reviewing PDF output styling.

Content Rendering

The preview renders the following DITA elements with appropriate styling:

- **Structure:** Sections, paragraphs, titles, and nested content
- **Lists:** Ordered, unordered, and definition lists
- **Tables:** Simple tables and complex tables with spanning
- **Code:** Code blocks and inline code with syntax highlighting
- **Notes:** Note, tip, warning, caution, and danger admonitions
- **Steps:** Task steps with numbering and substeps
- **Images:** Referenced images with alt text

Conref Resolution

The preview resolves content references (conref) and displays the referenced content inline. This lets you see your complete document with all reused content in place.

Example:

```
<p conref="shared/common.dita#common/legal-
notice"/>
```

In the preview, this shows the actual content from the referenced element, not just a placeholder.

Keyref Resolution

Key references are resolved using the current key space. If you have key definitions in your map, the preview shows the resolved values.

Preview Controls

The preview panel includes a toolbar with these controls:

- **Refresh:** Manually refresh the preview
- **Toggle Scroll Sync:** Enable/disable scroll synchronization
- **Toggle Print Mode:** Switch between screen and print styling
- **Open in Browser:** Open the preview in your default browser

Limitations

The live preview is designed for quick content review and has some limitations compared to full DITA-OT output:

- Styling may differ from your DITA-OT configuration
- Some advanced features like relationship tables are not rendered
- Custom plugins and extensions are not applied

For final output review, use **DITA: Publish to HTML5** to generate full DITA-OT output.

Chapter 13

Map Visualizer

View your DITA map structure as an interactive tree diagram.

Overview

The Map Visualizer provides a graphical representation of your DITA map hierarchy. It displays the relationship between maps, chapters, and topics in an easy-to-navigate tree view.

Command: DITA: Show Map Visualizer

Opening the Visualizer

1. Open a `.ditamap` or `.bookmap` file
2. Open Command Palette (**Ctrl+Shift+P**)
3. Type `DITA: Show Map Visualizer`
4. The visualizer opens in a side panel

Tree Structure

The visualizer displays your map hierarchy with different node types:

Map Node	The root of the tree, representing the ditamap or bookmap file.
Chapter/Part Nodes	Major divisions in bookmaps, shown as expandable folders.
Topic References	Individual topic files referenced by the map.
Nested Maps	Submaps referenced via `<mapref>` appear as expandable subtrees.

Navigating the Tree

- **Click:** Select a node and open the corresponding file in the editor
- **Double-click:** Open the file and close the visualizer panel
- **Expand/Collapse:** Click the arrow icons to show or hide child nodes
- **Right-click:** Access context menu with additional options

Visual Indicators

The visualizer uses icons and colors to convey information:

- **File Icons:** Different icons for maps, topics, and other file types
- **Validation Status:** Red indicators show files with errors
- **Missing Files:** Broken references are highlighted

- **Current File:** The currently edited file is highlighted

Real-time Updates

The visualizer updates automatically when:

- You add or remove topicref elements from the map
- You create or delete topic files
- Validation status changes for any file
- You switch between maps

Use Cases

- **Content Audit:** Review your document structure at a glance
- **Navigation:** Quickly jump to any topic in a large documentation set
- **Validation Review:** Identify which files have errors
- **Structure Planning:** Visualize how content is organized

Chapter 14

Key Resolution

Use keys for indirect addressing and content reuse in your DITA projects.

Overview

DITA keys provide a powerful mechanism for indirect addressing. Instead of using direct file paths, you define keys in your map and reference them throughout your topics. This enables:

- Easy updates when file locations change
- Conditional content through key scopes
- Centralized management of reusable content
- Simplified localization workflows

Defining Keys

Keys are defined in DITA maps using `<keydef>` elements or the `keys` attribute on `<topicref>`.

Basic Key Definition:

```
<keydef keys="product-name">
    <topicmeta>
        <keywords>
            <keyword>DitaCraft</keyword>
        </keywords>
    </topicmeta>
</keydef>
```

Key with href:

```
<keydef keys="install-guide" href="topics/
installation.dita"/>
```

Key on topicref:

```
<topicref keys="getting-started" href="topics/
getting-started.dita"/>
```

Using keyref

Reference keys using the `keyref` attribute to pull in text content or create links.

Text Replacement:

```
<!-- In your topic -->
<p>Welcome to <ph keyref="product-name"/>!</p>
```

```
<!-- Renders as -->
<p>Welcome to DitaCraft!</p>
```

Link Creation:

```
<xref keyref="install-guide">Installation Guide</
xref>
```

Using conkeyref

The `conkeyref` attribute combines key resolution with content referencing. It first resolves the key to a file, then pulls content from a specific element.

```
<!-- Key definition -->
<keydef keys="shared-warnings" href="shared/
warnings.dita"/>

<!-- Usage -->
<note conkeyref="shared-warnings/safety-warning"/>
```

This resolves `shared-warnings` to `shared/warnings.dita`, then pulls the element with `id="safety-warning"`.

Key Space

DitaCraft builds a key space from your root map. The key space includes:

- All key definitions in the root map
- Keys from referenced submaps
- Keys inherited through key scopes

Key Precedence: When the same key is defined multiple times, the first definition wins. Keys defined earlier in the map take precedence.

Key Scopes

Key scopes allow you to define variations of the same content for different contexts.

```
<topicgroup keyscope="admin">
    <keydef keys="audience">

 <topicmeta><keywords><keyword>administrators</
keyword></keywords></topicmeta>
    </keydef>
    <topicref href="admin-guide.dita"/>
</topicgroup>

<topicgroup keyscope="user">
    <keydef keys="audience">
        <topicmeta><keywords><keyword>end users</
keyword></keywords></topicmeta>
    </keydef>
    <topicref href="user-guide.dita"/>
</topicgroup>
```

DitaCraft Key Support

DitaCraft provides full support for DITA keys:

- **Key Resolution:** Automatically builds key space from root maps with improved root map discovery
- **Navigation:** Ctrl+Click on keyref and conkeyref navigates to the key's target. Cross-file Go to Definition resolves key references to their source files.
- **Preview:** Live preview shows resolved key values
- **Validation:** Warns about undefined or missing keys
- **Caching:** Key space is cached with TTL and LRU eviction for performance, with automatic invalidation when map files change
- **Key Scope Support:** Full `@keyscope` attribute handling with scope-qualified keys (e.g., `admin.audience`)
- **Root Map Selection:** Explicitly set the root map for precise key space control via **DITA: Set Root Map**
- **LSP Integration:** The LSP server maintains its own key space service for instant keyref and conkeyref resolution in Go to Definition and document links

Chapter 15

DITAVAL Support

DitaCraft provides full editing support for DITAVAL conditional processing files.

Overview

DITAVAL files define conditional processing profiles that control which content is included, excluded, or flagged during DITA publishing. DitaCraft recognizes `.ditaval` files and provides IntelliSense to help you author them efficiently.

What is DITAVAL?

A DITAVAL file is an XML file that specifies filtering and flagging rules for conditional DITA content. It uses attribute-value pairs to control which content appears in the output based on conditions like audience, platform, or product.

Basic DITAVAL Example:

```
<val>
    <prop action="include" att="audience"
 val="admin"/>
    <prop action="exclude" att="audience"
 val="novice"/>
    <prop action="flag" att="platform"
 val="windows">
        <startflag imageref="icons/windows.png">
            <alt-text>Windows</alt-text>
        </startflag>
    </prop>
</val>
```

IntelliSense for DITAVAL

DitaCraft provides the following IntelliSense features for DITAVAL files:

Element Completion	Auto-completion for DITAVAL elements: `<val>`, `<prop>`, `<revprop>`, `<startflag>`, `<endflag>`, and `<alt-text>`.
Attribute Completion	Suggestions for attributes on each element, including `action`, `att`, `val`, and `imageref`.
Attribute Value Completion	Predefined values for the `action` attribute (`include`, `exclude`,

passthrough, flag) and standard DITA conditional attributes for the att attribute (audience, platform, product, otherprops, props, rev).

DITAVAL Elements Reference

Element	Description
<val>	Root element of a DITAVAL file
<prop>	Defines a filtering or flagging rule based on an attribute-value pair
<revprop>	Defines a rule for revision-based filtering or flagging
<startflag>	Specifies a flag image or text inserted before flagged content
<endflag>	Specifies a flag image or text inserted after flagged content
<alt-text>	Alternative text for a flag image

Using DITAVAL with DITA-OT

To apply a DITAVAL profile when publishing, pass it as a filter argument to DITA-OT. In DitaCraft, you can add it to the ditacraft.ditaOtArgs setting:

```
"ditacraft.ditaOtArgs": ["--
filter=myprofile.ditaval"]
```

This applies the conditional processing rules from your DITAVAL file during publishing, filtering and flagging content as specified.

Chapter 16

Cross-Reference Validation

DitaCraft validates cross-file references including href, conref, keyref, and conkeyref attributes to catch broken links before publishing.

Overview

Cross-reference validation checks that all references in your DITA content point to valid targets. This includes file paths, topic IDs, element IDs, and key definitions. Broken references are reported as diagnostics so you can fix them before publishing.

Cross-reference validation runs automatically as part of the LSP diagnostics pipeline alongside XML well-formedness and DTD validation.

What Gets Validated

href Attributes	Validates that the target file exists. If the href includes a fragment identifier (e.g., `file.dita#topic_id/element_id`), the topic ID and element ID are also checked. External URLs (`http://`, `https://`) and references with `scope="external"` are skipped.
conref Attributes	Validates that the referenced file exists and that the target topic and element IDs are present in the target file. Same-file conrefs (starting with `#`) are validated against IDs within the current document.
keyref Attributes	Validates that the referenced key is defined in the key space. If the key has an href target, that target file is also checked.
conkeyref Attributes	Validates the key portion against the key space and, if present, the element ID portion against the target file.

Diagnostic Codes

Cross-reference validation produces the following diagnostic codes:

Code	Severity	Description
`DITA-XREF-001`	Warning	Target file not found
`DITA-XREF-002`	Warning	Topic ID not found in target file
`DITA-XREF-003`	Warning	Element ID not found in target file
`DITA-KEY-001`	Warning	Undefined key (not in key space)
`DITA-KEY-002`	Information	Key is defined but has no href target
`DITA-KEY-003`	Warning	Key target file is missing the referenced element ID

Configuration

Cross-reference validation is enabled by default. To disable it:

```
{
    "ditacraft.crossRefValidationEnabled": false
}
```

When disabled, only XML well-formedness, DTD, and DITA structure validation are performed.

Chapter 17

DITA Rules Engine

DitaCraft includes 35 Schematron-equivalent validation rules organized in four categories to enforce DITA best practices and specification requirements, including 10 DITA 2.0-specific rules.

Overview

The DITA Rules Engine validates your content against a set of rules derived from the DITA specification and authoring best practices. These rules go beyond DTD validation to catch semantic issues such as missing accessibility attributes, deprecated elements, and incorrect attribute usage.

Rules are version-aware: DitaCraft automatically detects your DITA version (from `@DITAArchVersion` or DOCTYPE) and only applies rules relevant to that version.

Rule Categories

Mandatory	Rules required by the DITA specification. These catch constraint violations that would cause processing errors. Examples: `role="other"` requires `@otherrole`, deprecated `<indextermref>` element usage.
Recommendation	Best practice rules recommended by the DITA specification. Examples: adding short descriptions, using proper list structures, following naming conventions.
Authoring	Content quality rules for consistent and well-structured documentation. Examples: title length limits, nesting depth, content organization.
Accessibility	Rules ensuring your content is accessible to all users. Examples: `<image>` elements should have `<alt>` text, deprecated `alt` attribute should be converted to an `<alt>` element.

Key Rules

Some commonly encountered rules include:

Code	Category	Description
`DITA-SCH-001`	Mandatory	`role="other"` requires `@otherrole` attribute
`DITA-SCH-002`	Mandatory	`<note type="other">` requires `@othertype` attribute
`DITA-SCH-003`	Mandatory	`<indextermref>` is deprecated
`DITA-SCH-004`	Mandatory	`collection-type` not allowed on `<reltable>` or `<relcolspec>`
`DITA-SCH-011`	Accessibility	Deprecated `alt` attribute on `<image>` — use `<alt>` element instead
`DITA-SCH-030`	Accessibility	Missing `<alt>` element for `<image>`
`DITA-SCH-040`	Mandatory	Nested `<xref>` not allowed
`DITA-SCH-044`	Recommendation	Deprecated role values (`sample`, `external`)
`DITA-SCH-046`	Recommendation	Elements with `<title>` should have `@id`

Quick Fixes

Several DITA rules offer automatic quick fixes via the lightbulb menu (**Ctrl +.**):

- **Add `@otherrole`:** Inserts the missing attribute when `role="other"` is detected
- **Remove deprecated `<indextermref>`:** Removes the deprecated element
- **Convert `alt` attribute to `<alt>` element:** Migrates the deprecated attribute syntax to the modern element form
- **Add missing `<alt>` to `<image>`:** Inserts an empty `<alt>` child element for accessibility

DITA 2.0 Rules

DitaCraft includes 10 rules specific to DITA 2.0 that detect removed elements and attributes. These rules only apply when the DITA version is detected as 2.0 (via `@DITAArchVersion` or DOCTYPE).

Code	Category	Description
DITA-SCH-050	Mandatory	<boolean> element removed in DITA 2.0
DITA-SCH-051	Mandatory	<indextermref> element removed in DITA 2.0
DITA-SCH-052	Mandatory	<object> removed — use <audio>, <video>, or <include>
DITA-SCH-053	Mandatory	Learning specializations removed in DITA 2.0
DITA-SCH-054	Accessibility	<audio> should have <fallback> element
DITA-SCH-055	Accessibility	<video> should have <fallback> element
DITA-SCH-056	Mandatory	@print attribute removed in DITA 2.0
DITA-SCH-057	Mandatory	@copy-to attribute removed in DITA 2.0
DITA-SCH-058	Mandatory	@navtitle attribute removed in DITA 2.0
DITA-SCH-059	Mandatory	@query attribute removed in DITA 2.0

Version Detection

DitaCraft automatically detects the DITA version of each document using these sources (in priority order):

1. The `ditacraft.ditaVersion` setting (if not set to `"auto"`)
2. The `@DITAArchVersion` attribute on the root element
3. The DOCTYPE public identifier (e.g., `-//OASIS//DTD DITA 2.0 Topic//EN`)
4. Defaults to DITA 1.3 if no version indicator is found

Version-specific rules are only applied when the detected version matches their applicability range.

Configuration

Enable or disable the entire rules engine:

```
{
    "ditacraft.ditaRulesEnabled": true
}
```

Select which rule categories to apply:

```
{
    "ditacraft.ditaRulesCategories": [
```

```
        "mandatory",
        "recommendation",
        "authoring",
        "accessibility"
    ]
}
```

For example, to run only mandatory and accessibility rules:

```
{
    "ditacraft.ditaRulesCategories": ["mandatory",
 "accessibility"]
}
```

Chapter 18

Profiling and Subject Scheme Validation

DitaCraft validates profiling attribute values against subject scheme constraints, ensuring controlled vocabularies are respected.

Overview

Subject scheme maps define controlled vocabularies for profiling attributes such as `audience`, `platform`, `product`, and `otherprops`. When a subject scheme is present in your workspace, DitaCraft validates that profiling attribute values use only the allowed values defined by the scheme.

How It Works

1. DitaCraft automatically discovers subject scheme maps in your workspace (or from the explicit root map)
2. The Subject Scheme Service parses the scheme and extracts the controlled values for each attribute, building hierarchy paths for grouped values
3. When you use profiling attributes in your DITA content, their values are checked against the allowed values
4. Invalid values are reported with a diagnostic that lists the valid options
5. Auto-completion for profiling attributes shows only allowed values with hierarchy grouping (e.g., `Platform > Linux > Ubuntu`) and default value preselection

Subject Hierarchy Grouping

When subject scheme definitions use nested `<subjectdef>` elements to define a hierarchy, DitaCraft displays this hierarchy in auto-completion suggestions. For example:

```
<subjectdef keys="os">
    <subjectdef keys="linux">
        <subjectdef keys="ubuntu"/>
        <subjectdef keys="fedora"/>
    </subjectdef>
    <subjectdef keys="windows"/>
</subjectdef>
```

When completing `platform` attribute values, you will see suggestions like:

- `ubuntu` — shown as `os > linux > ubuntu`
- `fedora` — shown as `os > linux > fedora`
- `windows` — shown as `os > windows`

If the scheme defines a default value, that value is preselected in the completion list.

Validated Attributes

The following profiling attributes are validated when controlled by a subject scheme:

- `audience`
- `platform`
- `product`
- `otherprops`
- `props`
- `deliveryTarget`

Attributes that are not constrained by a subject scheme are not validated — you can use any value freely.

Subject Scheme Example

A subject scheme map that constrains the `platform` attribute:

```
<subjectScheme>
    <subjectdef keys="platform-values">
        <subjectdef keys="windows"/>
        <subjectdef keys="linux"/>
        <subjectdef keys="macos"/>
    </subjectdef>
    <enumerationdef>
        <attributedef name="platform"/>
        <subjectdef keyref="platform-values"/>
    </enumerationdef>
</subjectScheme>
```

With this scheme active, using `platform="android"` would produce a warning because `android` is not in the allowed values list.

Diagnostic Codes

Code	Severity	Description
`DITA-PROF-001`	Warning	Attribute value not in controlled vocabulary. The diagnostic message lists the valid values.

Configuration

Subject scheme validation is enabled by default. To disable it:

```
{
    "ditacraft.subjectSchemeValidationEnabled":
 false
}
```

Chapter 19

Activity Bar Views

DitaCraft provides a dedicated sidebar in the VS Code Activity Bar with three tree views for project navigation, key management, and diagnostics.

Overview

The DitaCraft Activity Bar adds a dedicated icon to the VS Code sidebar. Clicking it reveals three specialized views that give you an at-a-glance overview of your DITA project:

- **DITA Explorer:** Browse all workspace maps with expandable hierarchy and type-specific icons
- **Key Space:** View all defined, undefined, and unused keys with usage locations
- **Diagnostics:** Centralized DITA validation issues grouped by file or severity

To open the DitaCraft sidebar, click the DitaCraft icon in the Activity Bar on the left side of VS Code.

Additionally, the VS Code status bar shows the current **root map** status. Click it to set or change the root map for key resolution and cross-reference validation. See Root Map Management on page 71 for details.

Views at a Glance

DITA Explorer	Shows all `.ditamap` and `.bookmap` files in your workspace as expandable trees. Each map displays its topic hierarchy with icons for chapters, topics, keydefs, and other element types. Click any item to open the referenced file. Right-click for context menu actions such as validate, publish, or show the map visualizer.
Key Space	Lists all DITA keys organized into three groups: **Defined Keys** (used and defined), **Undefined Keys** (used but not defined in any map), and **Unused Keys** (defined but never referenced). Expand any key to see where it is used across your workspace.
Diagnostics	Aggregates all DITA-related validation issues from the Problems

panel. Switch between grouping by file or by severity. Click any issue to navigate directly to its location in the editor.

Welcome Content

When a view has no data to display, DitaCraft shows helpful welcome messages:

- **DITA Explorer (no maps):** Suggests creating a new map or bookmap
- **DITA Explorer (no folder):** Suggests opening a folder containing DITA files
- **Key Space (empty):** Explains that keys will appear once maps with key definitions are found
- **Diagnostics (clean):** Confirms that no DITA validation issues were found

Refreshing Views

All three views auto-refresh when relevant files change. You can also manually refresh each view using the refresh button in the view title bar or the corresponding commands:

- **DITA: Refresh DITA Explorer**
- **DITA: Refresh Key Space**
- **DITA: Refresh Diagnostics**
- **DITA: Set Root Map** — Set an explicit root map for the project
- **DITA: Clear Root Map (Auto-Discover)** — Return to automatic root map discovery

Chapter 20

DITA Explorer

Browse all workspace DITA maps with an expandable hierarchy tree in the Activity Bar sidebar.

Overview

The DITA Explorer is the primary view in the DitaCraft Activity Bar. It discovers all `.ditamap` and `.bookmap` files in your workspace and displays each one as an expandable tree showing its full topic hierarchy.

Command: DITA: Refresh DITA Explorer

Opening the DITA Explorer

1. Click the DitaCraft icon in the Activity Bar (left sidebar)
2. The DITA Explorer view appears at the top of the sidebar
3. All workspace maps are listed as root nodes

Tree Structure

Each map is displayed as a tree with type-specific icons:

Map	Root-level ditamap or bookmap files, shown with a hierarchy icon. These are always the top-level nodes.
Chapter	Chapter elements from bookmaps, shown with a book icon.
Part	Part elements from bookmaps, shown with a folder icon.
Appendix	Appendix elements, shown with a file-add icon.
Topic Reference	Standard `<topicref>` elements, shown with a file icon.
Key Definition	`<keydef>` elements, shown with a key icon.
Missing File	References to files that do not exist are shown with a warning icon.

Nested maps referenced via `<mapref>` are automatically expanded as subtrees. Circular references are detected and marked.

Navigation and Actions

- **Click:** Open the referenced file in the editor

- **Expand/Collapse:** Click the arrow to show or hide child elements
- **Context Menu:** Right-click a node for additional actions:
  - **Validate** - Run validation on the file
  - **Publish** - Publish the file (maps only)
  - **Show Map Visualizer** - Open the Map Visualizer panel (maps only)

Each item shows the `href` value as a description next to the label, and a tooltip with type, href, keys, and status information.

Validation Badges

Files in the DITA Explorer show validation status badges from the File Decoration Provider:

- **Error badge:** Red indicator with error count for files with validation errors
- **Warning badge:** Yellow indicator with warning count for files with warnings only
- **No badge:** Clean files with no validation issues

Badges update automatically as diagnostics change.

Auto-Refresh

The DITA Explorer watches for changes to `.ditamap`, `.bookmap`, and `.dita` files. When files are created, modified, or deleted, the tree refreshes automatically after a 500ms debounce period to avoid excessive rebuilds during batch operations.

To manually refresh, click the refresh button in the view title bar or run **DITA: Refresh DITA Explorer** from the Command Palette.

Chapter 21

Key Space View

Browse all DITA keys in your workspace organized by status: defined, undefined, and unused.

Overview

The Key Space view in the DitaCraft Activity Bar shows every DITA key across your workspace, grouped by its resolution status. This helps you identify missing key definitions, find unused keys, and navigate to key usages.

Command: DITA: Refresh Key Space

Key Groups

Keys are organized into three groups:

Defined Keys	Keys that are both defined in a map (via `<keydef>` or `<topicref keys="...">`) and referenced somewhere in your DITA content (via `keyref` or `conkeyref` attributes). These are healthy keys that resolve correctly.
Undefined Keys	Keys that are referenced in your content but have no matching definition in any workspace map. These typically indicate a missing `<keydef>` element or a typo in a `keyref` attribute. Shown with a warning icon.
Unused Keys	Keys that are defined in a map but never referenced anywhere in your DITA content. These may be candidates for cleanup or may indicate content that needs to use the key.

Navigating Keys and Usages

- **Click a key:** Navigate to the map file where the key is defined
- **Expand a key:** See all files that reference this key via `keyref` or `conkeyref`
- **Click a usage:** Navigate directly to the usage location in the editor, with the reference highlighted

Each key item shows additional details:

- **Description:** The target filename for keys with an `href`, or "(inline)" for keys with inline content
- **Tooltip:** Key name, target file, source map, and scope
- **Usage description:** Reference type (`keyref` or `conkeyref`) and line number

How Keys Are Collected

DitaCraft builds the key space by:

1. Finding all `.ditamap` and `.bookmap` files in the workspace
2. Parsing each map to extract `<keydef>` and key-bearing `<topicref>` elements
3. Scanning all `.dita`, `.ditamap`, and `.bookmap` files (up to 500) for `keyref` and `conkeyref` attributes
4. Comparing definitions against usages to classify each key

For `conkeyref` values like `conkeyref="my-key/element-id"`, only the key name before the slash is extracted.

Auto-Refresh

The Key Space view watches for changes to `.ditamap` and `.bookmap` files. When map files are created, modified, or deleted, the view refreshes after a 1000ms debounce period. Key space resolution can be expensive for large workspaces, so the debounce is longer than other views.

To manually refresh, click the refresh button in the view title bar or run **DITA: Refresh Key Space** from the Command Palette.

Use Cases

- **Key Audit:** Review which keys are defined, used, and missing
- **Cleanup:** Identify unused keys that can be removed from maps
- **Debugging:** Find undefined keys causing unresolved references
- **Navigation:** Quickly jump to key definitions or usages across files

Chapter 22

Diagnostics View

View all DITA validation issues in a centralized tree with grouping by file or severity.

Overview

The Diagnostics view in the DitaCraft Activity Bar aggregates all DITA-related validation issues from across your workspace. Unlike the built-in Problems panel, this view filters specifically for DITA diagnostics and offers two grouping modes for easier triage.

Command: DITA: Refresh Diagnostics

Grouping Modes

Switch between two ways of organizing diagnostics using the view title bar buttons:

Group by File	Issues are grouped under their parent file. Each file group shows the total issue count and a summary of errors and warnings. The file icon color reflects the highest severity (red for errors, yellow for warnings). Click the **Group by File** button in the view title bar to activate.
Group by Severity	Issues are grouped into four severity categories: Errors, Warnings, Information, and Hints. Each group shows its count. Errors and warnings are expanded by default; information and hints are collapsed. Click the **Group by Severity** button in the view title bar to activate.

Diagnostic Items

Each diagnostic item displays:

- **Severity icon:** Error (red circle), Warning (yellow triangle), Information (blue info), or Hint (outline circle)
- **Message:** The validation issue description
- **Line number:** Shown as a description (e.g., "line 12")
- **Tooltip:** Full message, source (e.g., `dita`, `dita-lsp`, `ditacraft-keys`), and line number

Click any diagnostic item to open the file and jump to the exact location of the issue.

Diagnostic Sources

The Diagnostics view only shows issues from DITA-related sources:

- `dita` - Client-side DITA validation
- `dita-lsp` - Language Server validation (XML, structure, IDs)
- `ditacraft-keys` - Key reference diagnostics

Diagnostics from other extensions (e.g., spell checkers, generic XML linters) are excluded to keep the view focused on DITA-specific issues.

Auto-Refresh

The Diagnostics view refreshes automatically whenever validation results change (e.g., after editing a file, saving, or opening a new document). Updates are debounced at 300ms to batch rapid changes.

To manually refresh, click the refresh button in the view title bar or run **DITA: Refresh Diagnostics** from the Command Palette.

Clean State

When no DITA validation issues are found, the view shows a welcome message confirming that your workspace is clean. This is the ideal state after resolving all validation issues.

Chapter 23

Root Map Management

Explicitly set a root map for your project to control key resolution, cross-reference validation, and subject scheme discovery.

Overview

DitaCraft uses a root map to build the key space, resolve cross-references, and discover subject scheme constraints. By default, DitaCraft auto-discovers root maps in your workspace. Starting with v0.6.2, you can explicitly set a root map for precise control over key resolution and validation behavior.

Automatic Discovery

When no explicit root map is set, DitaCraft automatically discovers maps in your workspace using the following strategy:

1. Scans for `.ditamap` and `.bookmap` files
2. Selects the most likely root map based on structure and references
3. Builds the key space from the discovered map

Auto-discovery works well for projects with a single root map. For complex projects with multiple maps, setting an explicit root map is recommended.

Setting a Root Map

To explicitly set the root map:

1. Open the Command Palette (**Ctrl+Shift+P**)
2. Run **DITA: Set Root Map**
3. Select the desired map file from the list of available maps

The selected root map is stored in your workspace settings (`ditacraft.rootMap`) and persists across sessions.

Clearing the Root Map

To return to automatic discovery:

1. Open the Command Palette (**Ctrl+Shift+P**)
2. Run **DITA: Clear Root Map (Auto-Discover)**

This removes the explicit root map setting and DitaCraft reverts to auto-discovery mode.

Status Bar Indicator

The current root map status is displayed in the VS Code status bar at the bottom of the window. It shows:

- **Map name:** When an explicit root map is set, the filename is displayed
- **Auto:** When DitaCraft is using automatic map discovery

Click the status bar item to quickly change the root map.

Configuration

The root map setting can also be configured directly in `settings.json`:

```
{
    "ditacraft.rootMap": "maps/master.ditamap"
}
```

The path is relative to the workspace root. Leave empty or omit for automatic discovery.

What the Root Map Affects

The root map determines the scope for:

- **Key Space:** All `keyref` and `conkeyref` resolution uses keys defined in the root map hierarchy
- **Cross-Reference Validation:** Key-based references are validated against the root map's key space
- **Subject Scheme Discovery:** Subject scheme maps referenced from the root map define profiling constraints
- **DITA Explorer:** The root map hierarchy is used for the tree view
- **Key Space View:** Shows defined, undefined, and unused keys from the root map

Chapter 24

Localization (i18n)

DitaCraft supports localized diagnostic messages, currently available in English and French.

Overview

Starting with version 0.6.2, DitaCraft localizes diagnostic messages produced by the DITA Rules Engine, cross-reference validation, profiling validation, and structural validation. The language is detected automatically from your VS Code display language setting.

Supported Languages

Language	Locale Code	Status
English	`en`	Default / Fallback
French	`fr`	Full translation (67 messages)

How It Works

1. When the LSP server starts, it reads the locale from VS Code's `InitializeParams.locale` setting
2. The corresponding message bundle (`en.json` or `fr.json`) is loaded
3. All diagnostic messages use the localized strings with placeholder interpolation
4. If a translation is missing for a key, the English fallback is used automatically

What Is Localized

The following diagnostic messages are localized:

- **XML validation:** Syntax errors, well-formedness issues
- **DITA structure validation:** Missing DOCTYPE, missing ID, missing title, empty elements, duplicate IDs
- **Cross-reference validation:** Broken links, missing files, undefined keys
- **DITA Rules Engine:** All 35 rules (mandatory, recommendation, authoring, accessibility)
- **Profiling validation:** Invalid attribute values
- **DTD validation:** Catalog-based DTD conformance messages

Changing the Language

DitaCraft follows your VS Code display language. To change it:

1. Open the Command Palette (**Ctrl+Shift+P**)
2. Type `Configure Display Language`
3. Select your preferred language (e.g., `fr` for French)
4. Restart VS Code

DitaCraft diagnostics will automatically appear in the selected language on the next server start.

Part III

Configuration

Topics:

- Settings Overview
- General Settings
- Validation Settings
- Publishing Settings
- Preview Settings

Chapter 25

Settings Overview

Configure DitaCraft to match your workflow and preferences.

Accessing Settings

DitaCraft settings can be accessed in several ways:

1. Open VS Code Settings (**Ctrl+,** or **Cmd+,**)
2. Search for `ditacraft`
3. Or edit `settings.json` directly

Settings Levels

VS Code supports multiple settings levels:

User Settings	Apply to all VS Code instances. Stored in your user profile.
Workspace Settings	Apply only to the current workspace. Stored in `.vscode/settings.json`.
Folder Settings	Apply to a specific folder in a multi-root workspace.

Workspace settings override user settings, allowing project-specific configuration.

Settings Categories

DitaCraft settings are organized into these categories:

Category	Description
General	DITA-OT path and basic configuration
Validation	Validation engine and auto-validation settings
Publishing	Output directory and publishing options
Preview	Live preview behavior and styling

Quick Settings Reference

Setting	Default	Description
`ditacraft.ditaOtPath`	`""`	Path to DITA-OT installation
`ditacraft.validationEngine`	`"typesxml"`	Validation engine to use
`ditacraft.autoValidate`	`true`	Enable automatic validation
`ditacraft.outputDirectory`	`"out"`	Output directory for publishing
`ditacraft.previewScrollSync`	`true`	Enable preview scroll sync
`ditacraft.ditaRulesEnabled`	`true`	Enable DITA Rules Engine (35 rules)
`ditacraft.crossRefValidationEnabled`	`true`	Enable cross-reference validation
`ditacraft.ditaVersion`	`"auto"`	Override DITA version detection
`ditacraft.rootMap`	`""`	Explicit root map path

Example settings.json

```
{
    "ditacraft.ditaOtPath": "C:/DITA-OT-4.2.1",
    "ditacraft.validationEngine": "typesxml",
    "ditacraft.autoValidate": true,
    "ditacraft.outputDirectory": "out",
    "ditacraft.previewScrollSync": true,
    "ditacraft.ditaRulesEnabled": true,
    "ditacraft.ditaRulesCategories": ["mandatory",
 "recommendation", "authoring", "accessibility"],
    "ditacraft.crossRefValidationEnabled": true,
    "ditacraft.ditaVersion": "auto",
    "ditacraft.rootMap": ""
}
```

Chapter 26

General Settings

Configure DITA-OT path and basic DitaCraft settings.

ditacraft.ditaOtPath

Type: String

Default: `""` (empty)

The path to your DITA Open Toolkit installation directory. This setting is required for publishing features.

Example Values:

- Windows: `"C:/DITA-OT-4.2.1"`
- macOS: `"/Users/username/dita-ot-4.2.1"`
- Linux: `"/opt/dita-ot-4.2.1"`

Tip: Use forward slashes (/) in paths, even on Windows. VS Code handles path conversion automatically.

How to Set:

1. Run **DITA: Configure DITA-OT Path** from Command Palette, or
2. Set the value manually in settings.json

Verification:

DitaCraft verifies the path by checking for:

- `bin/dita` (Unix) or `bin/dita.bat` (Windows)
- `lib/` directory with required JAR files

Java Configuration

DITA-OT requires Java 11 or higher. DitaCraft uses the following to locate Java:

1. `JAVA_HOME` environment variable
2. Java in system PATH

If Java is not found, publishing will fail with a "Java not found" error.

Setting JAVA_HOME:

```
# Windows (PowerShell)
$env:JAVA_HOME = "C:\Program Files\Java\jdk-17"

# macOS/Linux
export JAVA_HOME=/usr/lib/jvm/java-17-openjdk
```

Workspace Detection

DitaCraft automatically detects workspace context:

- **Single Folder:** Uses the folder as the workspace root
- **Multi-root Workspace:** Uses the folder containing the current file
- **No Workspace:** Uses the directory of the current file

This affects relative path resolution for href, conref, and key definitions.

Chapter 27

Validation Settings

Configure validation engine, automatic validation, cross-reference checking, DITA rules, and profiling validation.

ditacraft.validationEngine

Type: String (enum)

Default: `"built-in"`

Options:

`"built-in"`	Lightweight validation with content model checking. Fast and suitable for most DITA authoring workflows. The default engine.
`"typesxml"`	Full DTD validation using TypesXML parser. Provides 100% W3C conformance with complete DTD support. Best for strict compliance checking.
`"xmllint"`	External validation using libxml2's xmllint tool. Requires xmllint to be installed on your system. Provides excellent performance and accuracy.

Choosing an Engine:

Use Case	Recommended Engine
General DITA authoring	`built-in` (default)
Strict DTD conformance required	`typesxml` or `xmllint`
Very large files (1000+ elements)	`built-in` or `xmllint`
Quick syntax checking only	`built-in`

ditacraft.autoValidate

Type: Boolean

Default: `true`

When enabled, DitaCraft validates DITA files automatically:

- When a file is opened
- When a file is saved
- After typing stops (with 500ms debounce)

Set to `false` to disable automatic validation. You can still validate manually using **Ctrl+Shift+V**.

When to Disable:

- Working with very large files that cause performance issues
- Editing files that are intentionally incomplete
- When you prefer manual validation control

ditacraft.maxNumberOfProblems

Type: Number

Default: `100`

The maximum number of diagnostic problems reported per file. Increase this value if you work with very large files and need to see all issues. Decrease it if the Problems panel becomes overwhelming.

ditacraft.ditaRulesEnabled

Type: Boolean

Default: `true`

Enable or disable the DITA Rules Engine. When enabled, DitaCraft validates your content against 35 Schematron-equivalent rules covering specification compliance, best practices, and accessibility (including 10 DITA 2.0-specific rules). Set to `false` to disable all DITA rules validation.

ditacraft.ditaRulesCategories

Type: Array of strings

Default: `["mandatory", "recommendation", "authoring", "accessibility"]`

Select which DITA rule categories to apply. Available categories:

- `"mandatory"` — Rules required by the DITA specification
- `"recommendation"` — Best practices recommended by the DITA specification
- `"authoring"` — Content quality and consistency rules
- `"accessibility"` — Accessibility compliance rules

For example, to run only mandatory and accessibility rules:

```
{
    "ditacraft.ditaRulesCategories": ["mandatory",
 "accessibility"]
}
```

ditacraft.crossRefValidationEnabled

Type: Boolean

Default: `true`

Enable or disable cross-reference validation. When enabled, DitaCraft checks that href, conref, keyref, and conkeyref attributes point to valid targets. Set to `false` if you are working with incomplete content where references are expected to be unresolved.

ditacraft.subjectSchemeValidationEnabled

Type: Boolean

Default: `true`

Enable or disable subject scheme validation. When enabled and a subject scheme map is present in the workspace, DitaCraft validates that profiling attribute values (audience, platform, product, etc.) match the controlled vocabulary defined by the scheme. Set to `false` to allow any profiling attribute values.

ditacraft.ditaVersion

Type: String (enum)

Default: `"auto"`

Options: `"auto"`, `"1.0"`, `"1.1"`, `"1.2"`, `"1.3"`, `"2.0"`

Override the auto-detected DITA version for validation. When set to `"auto"` (default), DitaCraft detects the version from `@DITAArchVersion` or DOCTYPE. Set an explicit version to force version-specific rules (e.g., `"2.0"` to enable DITA 2.0 rules for all documents).

ditacraft.schemaFormat

Type: String (enum)

Default: `"dtd"`

Options: `"dtd"`, `"rng"`

Schema format for validation. The default `"dtd"` uses standard DTD validation. Set to `"rng"` to enable RelaxNG schema validation using `salve-annos`. Requires `ditacraft.rngSchemaPath` to be set.

ditacraft.rngSchemaPath

Type: String

Default: `""`

Path to the directory containing RNG schema files (e.g., a DITA-OT's schema directory). Required when `ditacraft.schemaFormat` is set to `"rng"`. RNG schemas are compiled to JSON and cached for performance (up to 20 grammars).

ditacraft.rootMap

Type: String

Default: `""`

Explicit root map path (relative to workspace root). Leave empty for automatic root map discovery. Set via the **DITA: Set Root Map** command or directly in settings.

The root map determines the key space for keyref/conkeyref resolution, cross-reference validation scope, and subject scheme discovery. See Root Map Management on page 71 for details.

Rate Limiting

DitaCraft includes built-in rate limiting to prevent performance issues:

- **Limit:** 10 validation requests per second per file
- **Behavior:** Excess requests are silently skipped during auto-validation
- **Manual Validation:** Shows a warning if rate limited

Rate limiting cannot be configured but ensures stable performance during rapid editing.

Setting Up xmllint

To use the `xmllint` engine, install libxml2:

Windows:

1. Download from zlatkovic.com
2. Add the bin directory to your PATH

macOS:

```
brew install libxml2
```

Linux (Debian/Ubuntu):

```
sudo apt-get install libxml2-utils
```

Verification:

```
xmllint --version
```

Chapter 28

Publishing Settings

Configure output directory and DITA-OT publishing options.

ditacraft.outputDirectory

Type: String

Default: `"out"`

The directory where published output is generated. Can be an absolute path or relative to the workspace root.

Examples:

- `"out"` - Creates `{workspace}/out/`
- `"build/output"` - Creates `{workspace}/build/output/`
- `"C:/Documentation/Output"` - Uses absolute path

Output Structure:

Within the output directory, DitaCraft creates subdirectories for each format:

```
out/
├── html5/
│   ├── index.html
│   └── topics/
├── pdf/
│   └── document.pdf
└── xhtml/
    └── ...
```

Default Publishing Format

DitaCraft supports publishing to multiple formats via DITA-OT:

Format	Description	Requirements
html5	Modern responsive HTML5	None (built-in)
pdf	PDF via Apache FOP	FOP (included in DITA-OT)
xhtml	XHTML 1.0 Transitional	None (built-in)
eclipsehelp	Eclipse Help System	None (built-in)
htmlhelp	Microsoft HTML Help	HTML Help Compiler
markdown	Markdown output	None (built-in)

DITA-OT Build Parameters

DitaCraft passes the following parameters to DITA-OT during publishing:

Parameter	Value
`-i`	Input file (current map or topic)
`-f`	Output format (html5, pdf, etc.)
`-o`	Output directory

For advanced DITA-OT configuration, use a `build.xml` or `plugin.xml` file in your project.

Clean Output

By default, publishing overwrites existing output in the target directory. To ensure a clean build:

1. Delete the output directory manually, or
2. Use a different output directory for each build

Viewing Published Output

After publishing completes:

- **HTML5:** Open `out/html5/index.html` in a browser
- **PDF:** Open the generated PDF file in a PDF reader
- **VS Code:** Right-click HTML files and select "Open with Live Server"

Chapter 29

Preview Settings

Configure live preview behavior and appearance.

ditacraft.previewScrollSync

Type: Boolean

Default: `true`

Enables bidirectional scroll synchronization between the editor and preview panel.

Behavior when enabled:

- Scrolling in the editor scrolls the preview to the corresponding position
- Scrolling in the preview scrolls the editor to match
- Cursor position in editor highlights the corresponding preview element

When to Disable:

- Working with very long documents where sync causes performance issues
- When you prefer independent scrolling of editor and preview
- When editing and viewing different sections simultaneously

Theme Integration

The preview automatically adapts to VS Code's current theme:

- **Light Themes:** White background, dark text
- **Dark Themes:** Dark background, light text
- **High Contrast:** Enhanced contrast for accessibility

Preview styling follows VS Code's color theme to provide a consistent experience.

Preview Refresh Behavior

The preview refreshes automatically with the following characteristics:

- **Debounce:** 300ms delay after typing stops
- **Preservation:** Scroll position is preserved during refresh
- **Efficiency:** Only changed content is re-rendered when possible

Preview Panel Controls

The preview panel toolbar provides these controls:

Refresh Button	Manually refresh the preview content.

Scroll Sync Toggle	Enable or disable scroll synchronization.
Print Mode Toggle	Switch between screen and print styling. Print mode shows how content will appear when printed.
Open in Browser	Open the preview in your default web browser for full-screen viewing.

Content Resolution

The preview resolves the following content types:

Content Type	Resolution Behavior
conref	Fetches and displays referenced content inline
keyref	Resolves key to display value or creates link
conkeyref	Resolves key then fetches referenced content
images	Displays images from relative or absolute paths

Note: If a reference cannot be resolved, the preview shows a placeholder indicating the missing content.

Known Limitations

The live preview has some limitations compared to full DITA-OT output:

- Custom DITA-OT plugins are not applied
- Relationship tables are not rendered
- Some advanced table features may display differently
- PDF-specific formatting is not shown

For final output review, use the publishing commands to generate full DITA-OT output.

Appendix
A

Keyboard Shortcuts

Quick reference for all DitaCraft keyboard shortcuts.

Overview

DitaCraft provides keyboard shortcuts for common operations. This reference lists shortcuts for both Windows/Linux and macOS.

Validation

Windows/Linux	Mac	Command	Description
Ctrl+Shift+V	**Cmd+Shift+V**	DITA: Validate Current File	Validate the active DITA file against XML syntax and DTD

Publishing

Windows/Linux	Mac	Command	Description
Ctrl+Shift+B	**Cmd+Shift+B**	DITA: Publish (Select Format)	Publish with format selection dialog
Ctrl+Shift+H	**Cmd+Shift+H**	DITA: Preview HTML5	Open live HTML5 preview panel

Navigation

Windows/Linux	Mac	Action	Description
Ctrl+Click	**Cmd+Click**	Go to Definition	Navigate to href, conref, keyref, or conkeyref target
Ctrl+Alt+Click	**Cmd+Option +Click**	Open to Side	Open referenced file in split editor
F12	**F12**	Go to Definition	Navigate to the definition of the reference under cursor
Alt+F12	**Option+F12**	Peek Definition	Peek at definition inline without

Windows/Linux	Mac	Action	Description
			leaving current file
Shift+F12	**Shift+F12**	Find All References	Find all places where the current element is referenced
Alt+Left Arrow	**Ctrl+- (minus)**	Go Back	Return to previous location after navigation
Alt+Right Arrow	**Ctrl+Shift+- (minus)**	Go Forward	Move forward in navigation history

VS Code General Shortcuts

These VS Code shortcuts are useful when working with DITA files:

Windows/Linux	Mac	Action	Description
Ctrl+Shift+P	**Cmd+Shift+P**	Command Palette	Open Command Palette to access all DitaCraft commands
Ctrl+P	**Cmd+P**	Quick Open	Quickly open files by name
Ctrl+Shift+M	**Cmd+Shift+M**	Problems Panel	Show/hide validation errors and warnings
Ctrl+Shift+O	**Cmd+Shift+O**	Go to Symbol	Navigate to elements within the current file
Ctrl+G	**Ctrl+G**	Go to Line	Jump to a specific line number
Ctrl+/	**Cmd+/**	Toggle Comment	Comment/ uncomment XML content
Ctrl+Shift+[	**Cmd+Option+[**	Fold Region	Collapse the current XML element
Ctrl+Shift+]	**Cmd+Option+]**	Unfold Region	Expand the current XML element
Ctrl+K Ctrl+0	**Cmd+K Cmd+0**	Fold All	Collapse all foldable regions
Ctrl+K Ctrl+J	**Cmd+K Cmd+J**	Unfold All	Expand all folded regions

Windows/Linux	Mac	Action	Description
Ctrl+S	**Cmd+S**	Save	Save file (triggers auto-validation if enabled)
Ctrl+Z	**Cmd+Z**	Undo	Undo last edit
Ctrl+Shift+Z	**Cmd+Shift+Z**	Redo	Redo last undone edit

XML Editing Shortcuts

Windows/Linux	Mac	Action	Description
Ctrl+Space	**Ctrl+Space**	Trigger Suggestions	Show element and attribute auto-completion
Ctrl+Shift +Space	**Cmd+Shift +Space**	Parameter Hints	Show attribute value suggestions
Ctrl+.	**Cmd+.**	Quick Fix	Show available quick fixes for errors
F2	**F2**	Rename Symbol	Rename element ID across files
Ctrl+D	**Cmd+D**	Add Selection	Select next occurrence of current word
Ctrl+Shift+L	**Cmd+Shift+L**	Select All Occurrences	Select all occurrences of current word
Alt+Up/Down	**Option+Up/ Down**	Move Line	Move current line up or down
Shift+Alt+Up/ Down	**Shift+Option +Up/Down**	Copy Line	Duplicate current line above or below
Ctrl+Shift+K	**Cmd+Shift+K**	Delete Line	Delete the current line

Customizing Shortcuts

To customize keyboard shortcuts:

1. Press **Ctrl+K Ctrl+S** (Windows/Linux) or **Cmd+K Cmd+S** (Mac) to open Keyboard Shortcuts
2. Search for the command (e.g., "DITA: Validate")
3. Click the pencil icon to assign a new shortcut
4. Press your desired key combination

You can also edit `keybindings.json` directly for advanced customization.

Quick Cheat Sheet

Most frequently used DitaCraft shortcuts:

Action	**Windows/Linux**	**Mac**
Validate File	**Ctrl+Shift+V**	**Cmd+Shift+V**
Quick Fix (Code Actions)	**Ctrl+.**	**Cmd+.**
Publish	**Ctrl+Shift+B**	**Cmd+Shift+B**
Preview HTML5	**Ctrl+Shift+H**	**Cmd+Shift+H**
Navigate to Reference	**Ctrl+Click**	**Cmd+Click**
Auto-Complete	**Ctrl+Space**	**Ctrl+Space**
Hover Info	Mouse hover / **Ctrl+K Ctrl+I**	Mouse hover / **Cmd+K Cmd+I**
Command Palette	**Ctrl+Shift+P**	**Cmd+Shift+P**
Problems Panel	**Ctrl+Shift+M**	**Cmd+Shift+M**
Format Document	**Shift+Alt+F**	**Shift+Option+F**

Appendix
B

Auto-validation

A DitaCraft feature that automatically validates DITA files when they are opened, saved, or modified. Controlled by the `ditacraft.autoValidate` setting.

Related reference

Validation Settings on page 81

Appendix C

Bookmap

A specialized DITA map designed for book-oriented publications. Bookmaps support book-specific elements like chapters, parts, appendixes, frontmatter, and backmatter. File extension: `.bookmap`.

Related reference

File Creation Commands on page 25

Appendix
D

Built-in Engine

A lightweight validation engine bundled with DitaCraft that provides fast XML syntax checking and basic content model validation. Faster than TypesXML but with limited DTD support.

Related reference

Validation Settings on page 81

Appendix
E

OASIS XML Catalog

A standard mechanism for mapping PUBLIC identifiers (such as DOCTYPE declarations) to local file paths. DitaCraft uses an OASIS XML Catalog to resolve DITA DTD references to bundled DTD files, enabling offline validation without network access.

Appendix
F

Code Action

A quick fix or automated refactoring offered by the Language Server Protocol when a diagnostic issue is detected. In DitaCraft, code actions appear via the lightbulb icon or Ctrl+. shortcut and include fixes such as adding missing DOCTYPE, fixing duplicate IDs, adding accessibility attributes, and removing deprecated elements.

Related concepts

IntelliSense on page 33

DITA Rules Engine on page 57

Appendix
G

conkeyref

A DITA attribute that combines key resolution with content referencing. The key is first resolved to a file path, then the element ID is used to pull in specific content. Format: `conkeyref="keyname/elementid"`.

Related concepts

Key Resolution on page 49

Smart Navigation on page 37

Appendix
H

conref

Content reference. A DITA mechanism for reusing content by referencing an element in another file. The referenced content is pulled in during processing. Format: `conref="file.dita#topicid/elementid"`.

Related concepts

Smart Navigation on page 37

Appendix
I

Concept

A DITA topic type designed for conceptual information that helps users understand a subject. Concepts explain "what" and "why" rather than "how". Uses the `<concept>` root element with `<conbody>`.

Related reference

File Creation Commands on page 25

Appendix
J

Cross-Reference Validation

A validation feature that checks href, conref, keyref, and conkeyref attributes to ensure they point to valid targets. Cross-reference validation catches broken links, missing files, undefined keys, and invalid element IDs before publishing.

Related concepts

Cross-Reference Validation on page 55

Related reference

Validation Settings on page 81

Appendix
K

DITA

Darwin Information Typing Architecture. An XML-based standard for authoring, producing, and delivering technical documentation. DITA uses specialized topic types and maps to organize modular, reusable content.

Related concepts

Introduction to DitaCraft on page 13

Appendix
L

DITA 2.0

The latest major version of the DITA specification, which removes several deprecated elements (`<boolean>`, `<indextermref>`, `<object>`, learning specializations) and attributes (`@print`, `@copy-to`, `@navtitle`, `@query`), while adding new multimedia elements (`<audio>`, `<video>`, `<fallback>`). DitaCraft includes 10 DITA 2.0-specific validation rules.

Appendix
M

DITA-OT

DITA Open Toolkit. An open-source publishing engine that transforms DITA content into various output formats including HTML5, PDF, EPUB, and more. Required for DitaCraft publishing features.

Related reference

Publishing Commands on page 23

General Settings on page 79

Appendix
N

DITA Rules Engine

A validation engine that applies 18 Schematron-equivalent rules to enforce DITA specification requirements and best practices. Rules are organized in four categories: mandatory, recommendation, authoring, and accessibility. The engine is version-aware and only applies rules relevant to the detected DITA version.

Related concepts

DITA Rules Engine on page 57

Related reference

Validation Settings on page 81

Appendix
O

DITA Map

An XML file that defines the structure and organization of DITA topics. Maps specify which topics to include, their hierarchy, and relationships. File extension: `.ditamap`.

Related concepts

Map Visualizer on page 47

Related reference

File Creation Commands on page 25

Appendix
P

DTD

Document Type Definition. A set of rules that define the valid structure, elements, and attributes for an XML document. DitaCraft validates DITA files against DITA 1.3 DTDs.

Related concepts

Validation on page 41

Appendix
Q

href

Hypertext reference. A DITA attribute that specifies the location of a referenced resource. Used in `<topicref>`, `<xref>`, `<link>`, and other elements. Supports Ctrl+Click navigation in DitaCraft.

Related concepts

Smart Navigation on page 37

Appendix
R

Internationalization (i18n)

The process of designing software so that it can be adapted to various languages without engineering changes. DitaCraft supports localized diagnostic messages in English and French, with the language automatically detected from your VS Code display language setting.

Appendix
S

Keyboard Shortcuts

Key combinations that provide quick access to DitaCraft commands and VS Code features. Primary shortcuts include **Ctrl+Shift+V** (validate), **Ctrl+Shift+H** (preview), and **Ctrl+Click** (navigate).

Related reference

Keyboard Shortcuts on page 89

Appendix
T

keydef

Key definition. A DITA map element that defines a key and its associated resource or value. Keys enable indirect addressing and centralized management of reusable content.

Related concepts

Key Resolution on page 49

Appendix
U

keyref

Key reference. A DITA attribute that references a key defined in a map. When processed, the keyref is resolved to the key's target value or resource. Supports Ctrl+Click navigation in DitaCraft.

Related concepts

Key Resolution on page 49

Smart Navigation on page 37

Appendix
V

Key Scope

A DITA feature (via the `@keyscope` attribute) that enables different key definitions within different contexts of the same map. Key scopes allow the same key name to resolve to different targets depending on the context. DitaCraft supports scope-qualified key references (e.g., `admin.audience`).

Appendix
W

Key Space

The collection of all key definitions available in a DITA map and its submaps. DitaCraft builds the key space from the root map to resolve keyref and conkeyref attributes. The key space is cached for performance.

Related concepts

Key Resolution on page 49

Appendix
X

Live Preview

A DitaCraft feature that renders DITA content as HTML5 in real-time. The preview updates automatically as you edit and supports bidirectional scroll synchronization. Shortcut: **Ctrl+Shift+H**.

Related concepts

Live Preview on page 45

Related reference

Preview Settings on page 87

Appendix
Y

Profiling

A DITA mechanism for filtering and flagging content based on attribute values. Profiling attributes such as audience, platform, product, and otherprops allow conditional processing where content is included or excluded based on the target output. DitaCraft validates profiling values against subject scheme constraints when available.

Related concepts

Profiling and Subject Scheme Validation on page 61

DITAVAL Support on page 53

Appendix
Z

Map Visualizer

A DitaCraft feature that displays your DITA map hierarchy as an interactive tree diagram. Click any node to navigate to the corresponding file. The visualizer updates in real-time as you modify the map.

Related concepts

Map Visualizer on page 47

Appendix
AA

Rate Limiting

A DitaCraft protection mechanism that limits validation requests to prevent performance issues during rapid editing. Maximum 10 requests per second per file. Excess requests during auto-validation are silently skipped.

Related concepts

Validation on page 41

Related reference

Validation Commands on page 21

Appendix
AB

Reference

A DITA topic type designed for reference information such as API documentation, command syntax, or configuration parameters. Uses the `<reference>` root element with `<refbody>`.

Related reference

File Creation Commands on page 25

Appendix
AC

RNG (RelaxNG)

A schema language for XML that provides an alternative to DTD for defining document structure. DitaCraft supports optional RNG validation via the `salve-annos` library. Enable it by setting `ditacraft.schemaFormat` to `"rng"` and providing a path to RNG schema files.

Appendix
AD

Root Map

The top-level DITA map that defines the scope for key resolution, cross-reference validation, and subject scheme discovery. DitaCraft can auto-discover root maps or you can set one explicitly via the `DITA: Set Root Map` command. The root map determines which keys are available and which profiling constraints apply.

Appendix
AE

Scroll Sync

Bidirectional scroll synchronization between the editor and preview panel. When enabled, scrolling in either pane automatically scrolls the other to the corresponding position. Controlled by `ditacraft.previewScrollSync`.

Related concepts

Live Preview on page 45

Related reference

Preview Settings on page 87

Appendix
AF

Subject Scheme

A type of DITA map that defines controlled vocabularies for profiling attributes. Subject scheme maps specify allowed values for attributes such as audience, platform, product, and otherprops. DitaCraft uses subject schemes to validate that profiling attribute values conform to the defined vocabulary.

Related concepts

Profiling and Subject Scheme Validation on page 61

Appendix
AG

Smart Navigation

A DitaCraft feature that enables Ctrl+Click navigation on DITA reference attributes including href, conref, keyref, and conkeyref. Clicking a reference opens the target file and positions the cursor at the referenced element.

Related concepts

Smart Navigation on page 37

Related reference

Navigation Commands on page 27

Appendix
AH

Task

A DITA topic type designed for procedural information with step-by-step instructions. Tasks explain "how to" perform actions. Uses the `<task>` root element with `<taskbody>` containing `<steps>`.

Related reference

File Creation Commands on page 25

Appendix
AI

Topic

The basic unit of content in DITA. A topic is a standalone unit of information that addresses a single subject. DITA provides specialized topic types: concept, task, reference, and generic topic. File extension: `.dita`.

Related concepts

Introduction to DitaCraft on page 13

Related reference

File Creation Commands on page 25

Appendix
AJ

topicref

A DITA map element that references a topic to include in the publication. The `<topicref>` element uses the `href` attribute to specify the topic file and can be nested to create hierarchy.

Related concepts

Map Visualizer on page 47

Related reference

File Creation Commands on page 25

Appendix
AK

TypesXML

A pure TypeScript XML parser used by DitaCraft for DTD validation. TypesXML provides 100% W3C conformance and full DTD support without requiring external dependencies. The default and recommended validation engine.

Related concepts

Validation on page 41

Related reference

Validation Settings on page 81

Appendix
AL

Validation

The process of checking DITA content for XML well-formedness and DTD conformance. DitaCraft provides real-time validation with detailed error messages and multiple validation engine options.

Related concepts

Validation on page 41

Related reference

Validation Commands on page 21

Validation Settings on page 81

Appendix
AM

Validation Engine

A component that performs XML and DTD validation. DitaCraft supports three engines: TypesXML (default, full DTD support), built-in (lightweight, fast), and xmllint (external, industry-standard). Configured via `ditacraft.validationEngine`.

Related concepts

Validation on page 41

Related reference

Validation Settings on page 81

Appendix
AN

xmllint

A command-line XML validation tool from the libxml2 library. DitaCraft can use xmllint as an external validation engine. Provides excellent performance and accuracy but requires separate installation on your system.

Related reference

Validation Settings on page 81